

The main auxin biosynthesis pathway in *Arabidopsis*

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The phytohormone auxin plays critical roles in the regulation of plant growth and development. Indole-3-acetic acid (IAA) has been recognized as the major auxin for more than 70 y. Although several pathways have been proposed, how auxin is synthesized in plants is still unclear. Previous genetic and enzymatic studies demonstrated that both TRYPTOPHAN AMINOTRANSFERASE OF ARABIDOPSIS (TAA) and YUCCA (YUC) flavin monooxygenase-like proteins are required for biosynthesis of IAA during plant development, but these enzymes were placed in two independent pathways. In this article, we demonstrate that the TAA family produces indole-3-pyruvic acid (IPA) and the YUC family functions in the conversion of IPA to IAA in *Arabidopsis* (*Arabidopsis thaliana*) by a quantification method of IPA using liquid chromatography–electrospray ionization–tandem MS. We further show that YUC protein expressed in *Escherichia coli* directly converts IPA to IAA. Indole-3-acetaldehyde is probably not a precursor of IAA in the IPA pathway. Our results indicate that YUC proteins catalyze a rate-limiting step of the IPA pathway, which is the main IAA biosynthesis pathway in *Arabidopsis*.

plant hormone | metabolism

Auxin plays fundamental roles in plant growth and development. Auxin regulates cell division, cell expansion, cell differentiation, lateral root formation, flowering, and tropic responses (1). After the discovery of indole-3-acetic acid (IAA) in the 1930s, auxin has been virtually synonymous with IAA for more than 70 y. Recent studies demonstrated that IAA directly interacts with the F-box protein TIR1, and promotes the degradation of the Aux/IAA transcriptional repressors to activate diverse auxin responsive genes (2–4). Despite the importance of IAA in plants, IAA biosynthesis is not fully understood, most likely because of the existence of multiple pathways and functional redundancy of enzymes within the pathway (5, 6).

Genetic and biochemical studies indicated that tryptophan (Trp) is the main precursor for IAA in plants (5, 6). Alternatively, the Trp-independent pathway has been proposed for IAA biosynthesis, but a genetic basis for this pathway has not been defined (6–8). There are four proposed pathways for biosynthesis of IAA from Trp in plants: (i) the YUCCA (YUC) pathway, (ii) the indole-3-pyruvic acid (IPA) pathway, (iii) the indole-3-acetamide (IAM) pathway, and (iv) the indole-3-acetaldoxime (IAOx) pathway (previously called the CYP79B pathway), as shown in Fig. 1A (6, 9). Recent studies indicated that the IAOx pathway operates in relatively few plant species that have CYP79B family members to convert Trp to IAOx (9–13). IAOx was identified in *Arabidopsis*, but not from CYP79B-deficient mutants and several noncrucifer plants (9, 14, 15). The IAM pathway has been suggested to exist widely in plants, but it remains unclear exactly how IAM is produced (16). The conversion of IAM to IAA by *Arabidopsis* AMIDASE 1 (AMI1) has been demonstrated (17). The physiological significance of the IAM pathway in plants is under investigation.

The YUC pathway has been proposed as a common IAA biosynthetic pathway that produces auxin essential for embryo-

genesis, flower development, seedling growth, and vascular patterning (18–21). YUC genes have been identified ubiquitously in various plant species (22). In maize, a monocot-specific YUC-like protein SPARSE INFLORESCENCE 1 (SPI1) plays critical roles in vegetative and reproductive development (22). YUC family encode flavin monooxygenase-like proteins that catalyze a rate-limiting step in IAA biosynthesis (23). *Arabidopsis yuc1D* mutants, in which YUC1 is expressed under the control of cauliflower mosaic virus 35S promoter, show slightly increased IAA levels along with high-auxin phenotypes such as elongated hypocotyls, epinastic leaves, and enhanced apical dominance (23). *Arabidopsis* has 11 YUC genes, and yuc multiple KO mutants show severe auxin-deficient phenotypes (19, 20). YUC catalyzes the conversion of tryptamine (TAM) to *N*-hydroxy-TAM (HTAM) in vitro (23, 24). IAOx and indole-3-acetonitrile (IAN) were previously proposed as possible intermediates in the conversion of HTAM to IAA (23). However, our previous study indicated that IAOx and IAN are not common intermediates of IAA biosynthesis in plants (9). The underlying pathway from HTAM to IAA is still unknown.

More recent studies have isolated three *Arabidopsis* mutants—*shade avoidance 3*, *weak ethylene insensitive 8* (*wei8*), and *transport inhibitor response 2*—in which the TRYPTOPHAN AMINOTRANSFERASE OF ARABIDOPSIS 1 (*TAA1*) gene is disrupted (25–27). TAA1 mediates the conversion of Trp to IPA in the first step of the IPA pathway (Fig. 1A). TAA1 plays critical roles in embryogenesis, flower development, seedling growth, vascular patterning, lateral root formation, tropism, shade avoidance, and temperature-dependent hypocotyl elongation (25–27). There are two TAA1-related proteins—TAR1 and TAR2—in *Arabidopsis*. Double-KO mutants of *TAA1* and *TAR2* genes, *wei8 tar2*, showed severe growth defects caused by a significant reduction of IAA production in *Arabidopsis* (26). In maize, *VANISHING TASSEL 2* (*VT2*) gene has been identified to encode a grass-specific TAA1 orthologue required for vegetative and reproductive development (28). The pathway from IPA to IAA via indole-3-acetaldehyde (IAAld) by IPA DECARBOXYLASE (IPD) and ALDEHYDE OXIDASE (AO) has been proposed (5, 29, 30). However, IPD genes have not yet been identified in plants. There are four AO genes in *Arabidopsis*. It has been demonstrated that ARABIDOPSIS ALDEHYDE OXIDASE 1 (AAO1) can convert IAAld

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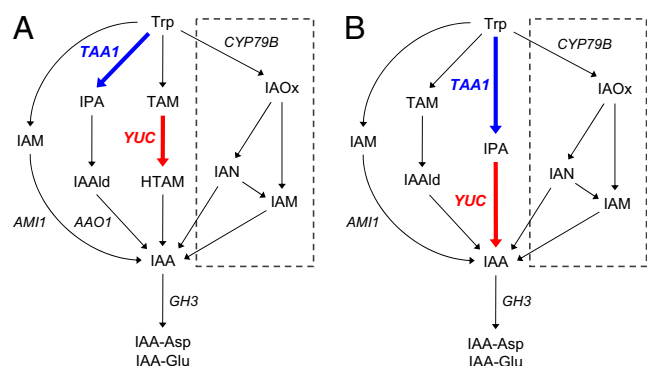


Fig. 1. Proposed IAA biosynthesis pathway in plants. (A) Previously proposed IAA biosynthesis pathway. (B) The IAA biosynthesis pathway proposed in the present study. The bold arrows indicate proposed functions of TAA1 and YUC, respectively. The IAOx pathway is illustrated in a dotted square. IAA-Asp and IAA-Glu are IAA metabolites investigated in this study.

to IAA (Fig. 1A) (31). AO family requires a molybdenum cofactor sulfuryase encoded by *ABA DEFICIENT 3* (*ABA3*) for its enzyme activity (32, 33). However, as *aba3*-deficient mutants do not show an apparent auxin-deficient phenotype, it is not clear whether the AO family actually participates in IAA biosynthesis in plants.

The IPA and YUC pathways have been proposed to independently produce IAA (Fig. 1A). However, the phenotypic similarities between TAA-deficient and YUC-deficient mutants suggested that TAA and YUC families possibly operate in the same auxin biosynthetic pathway (6, 8). A recent genetic study in maize led to the proposal that VT2 and SPI1, coorthologues of TAA and YUC, may function in the same IAA biosynthetic pathway, as there was no significant change in IAA levels between *vt2 spi1* double mutants and *vt2* single mutants (28).

Here, we provide genetic, enzymatic, and metabolite-based evidence that TAA and YUC families function in the same auxin biosynthetic pathway (Fig. 1B). YUC is implicated in the conversion of IPA to IAA in *Arabidopsis*. IAAld is probably not a precursor of IAA in the IPA pathway. We conclude that YUC family catalyzes a rate-limiting step of the IPA pathway that produces IAA essential for plant development.

Results

Synergistic Interaction Between TAA and YUC Families in IAA Biosynthesis. To investigate whether TAA and YUC families act in the same pathway, we generated estradiol (Est)-inducible *TAA1* overexpression plants in *Arabidopsis* WT (*TAA1ox*) and *yuc1D* (*TAA1ox yuc1D*), respectively. We predicted that co-overexpression of *TAA1* genes would enhance IAA biosynthesis in *yuc1D* mutants if TAA1 and YUC1 act in the same pathway. *TAA1ox* plants did not show apparent phenotypes relative to vector control plants (*pER8*) on Murashige–Skoog agar media containing Est (Fig. 2 A–C and Fig. S1). This observation strengthens the result of Tao et al. that TAA1 does not mediate a rate-limiting step in IAA biosynthesis (25). We found that the formation of adventitious and lateral roots was significantly enhanced in *TAA1ox yuc1D* plants relative to that in *yuc1D* mutants (Fig. 2 A, D, and E and Fig. S1). To determine if overexpression of *TAA1* enhances IAA biosynthesis in *yuc1D* mutants, we analyzed IAA levels in these mutants by liquid chromatography–electrospray ionization–tandem MS (LC-ESI-MS/MS). We also analyzed the levels of two IAA–amino acid conjugates, IAA–aspartate (IAA–Asp) and IAA–glutamate (IAA–Glu). IAA is metabolized to IAA–Asp, IAA–Glu, and other amino acid conjugates by the GH3 family for homeostatic regulation of auxin in plants (Fig. 1) (34). Hence, the GH3 family may greatly contribute to maintaining the level of IAA if excess amounts of IAA were produced in *TAA1ox yuc1D* mutants. As shown in Table 1,

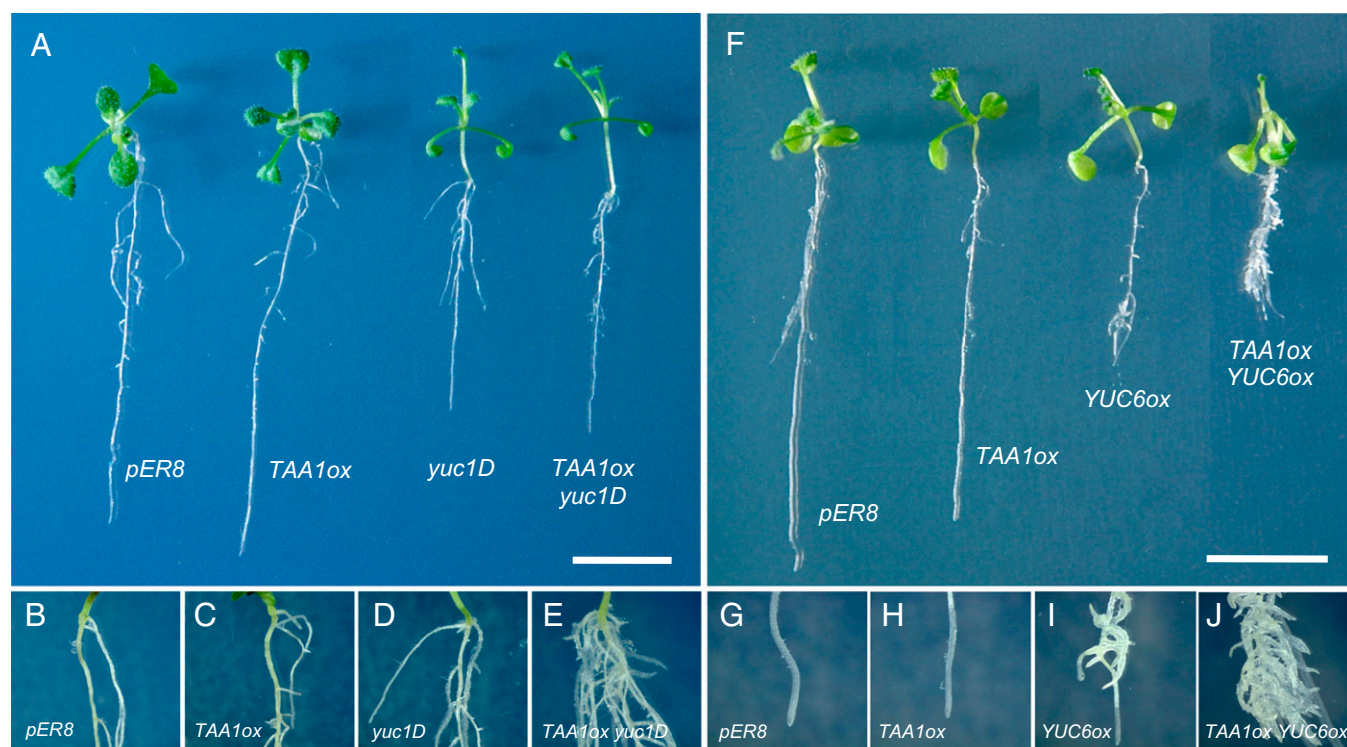


Fig. 2. Phenotypes of TAA1 and YUC overexpression plants in *Arabidopsis*. (A) Ten-day-old seedlings of *pER8*, *TAA1ox*, *yuc1D*, and *TAA1ox yuc1D* and (B–E) magnification of stem–root junctions (Est treatment for 5 d). (F) Est-treated 10-d-old seedlings of *pER8*, *TAA1ox*, *YUC6ox*, and *TAA1ox YUC6ox* and (G–J) magnification of root tip region (Est treatment for 5 d). (Scale bars: 1 cm.)

Table 1. IAA and IAA amino acid conjugate levels in seedlings of *TAA1*-, *YUC1*-, and *YUC6*-overexpressing plants

Plants	IAA and IAA metabolites (ng/gfw)		
	IAA	IAA-Asp	IAA-Glu
<i>pER8</i>	20.6 ± 1.7	ND	1.2 ± 0.5
<i>TAA1ox</i>	29.6 ± 2.1*	ND	1.1 ± 0.4
<i>yuc1D</i>	25.3 ± 3.4	ND	8.1 ± 1.6*
<i>TAA1ox yuc1D</i>	37.2 ± 4.0*	ND	18.7 ± 5.0*,†
<i>YUC6ox</i>	27.5 ± 2.0*	61.8 ± 16	28.9 ± 4.2*
<i>TAA1ox YUC6ox</i>	50.7 ± 2.0*,†	5,930 ± 175†	657 ± 125*,†

Four-day-old seedlings were transferred to Murashige-Skoog agar media containing Est (10 μM) and grown vertically for 4 d. ND, not detected. Values are mean ± SD, *n* = 3.

*Significantly different from *pER8* plants (*P* < 0.05, *t* test).

†Significantly different from either single overexpression line (*P* < 0.05, *t* test). In the case of IAA-Asp, significant difference from *YUC6ox* is shown.

IAA level increased slightly, but IAA-Asp and IAA-Glu levels did not change, in *TAA1ox* compared with that in *pER8*. In *yuc1D* mutants, IAA levels were not affected, but IAA-Glu levels increased by 6.8 times. We found that both IAA and IAA-Glu levels were 1.5 times and 2.3 times elevated, respectively, in *TAA1ox yuc1D* relative to that in *yuc1D* (Table 1). This suggests that GH3 family possibly metabolized excess amounts of IAA in these mutants. A significant increase in total levels of IAA and IAA-Glu in *TAA1ox yuc1D* relative to *yuc1D* indicates that *TAA1* and *YUC1* act synergistically to enhance IAA biosynthesis in *Arabidopsis*.

To further demonstrate the tandem action of TAA and YUC families in IAA biosynthesis, we generated Est-inducible *YUC6* overexpression plants (*YUC6ox*) and *TAA1 YUC6* cooverexpression plants (*TAA1ox YUC6ox*) in *Arabidopsis* (Fig. S1). We predicted that induction of both *TAA1* and *YUC6* genes would more efficiently enhance IAA biosynthesis relative to induction of *TAA1* gene in *yuc1D*, a weak allele of constitutive *YUC1* overexpression mutants. *YUC6ox* exhibited elongated hypocotyls and petioles, root growth inhibition, and enhanced lateral root and adventitious root formation like *yuc1D* on Murashige-Skoog agar media containing Est (Fig. 2 F, G, and I and Fig. S1). Similar to that observed in *TAA1 yuc1D*, adventitious roots and lateral roots were enhanced, but more strongly in *TAA1ox YUC6ox* cooverexpression plants (Fig. 2 F and H–J and Fig. S1). The level of IAA increased by only 1.8 times, but IAA-Asp and IAA-Glu levels were elevated by 96 and 23 times, respectively, in *TAA1ox YUC6ox* compared with *YUC6ox* (Table 1). These results indicate that TAA and YUC families are likely arranged in the same IAA biosynthesis pathway in *Arabidopsis*.

TAA Family Mainly Produces IPA from Trp in *Arabidopsis*. Enzymatic functions of TAA1 and YUC1/6 have been demonstrated by using their recombinant proteins in vitro (23–26), but their major functions may actually differ in plants. To complement our genetic evidence with a metabolite-based approach, we analyzed possible IAA precursors by using LC-ESI-MS/MS. IPA is an enzymatic reaction product of TAA1 in vitro. IPA is a relatively unstable IAA precursor and nonenzymatically converted to IAA in aqueous solution (35). To avoid the degradation of IPA during the purification, we immediately derivatized IPA with dinitrophenyl hydrazine (DNPH) to a stable hydrazone derivative (DNPH-IPA) in the crude extracts (Fig. S24). After purification, DNPH-IPA was further derivatized with diazomethane to methyl ester (DM-IPA), and analyzed using LC-ESI-MS/MS in the negative ion mode (Fig. S2 A–J).

By using this IPA analysis method, we tested if IPA is mainly produced from Trp in *Arabidopsis*. To selectively and efficiently

label IAA precursors in the Trp-dependent pathway with stable isotopes, Trp-auxotroph *trp1-1* mutants were supplemented with [¹³C₁₁, ¹⁵N₂]Trp in the liquid media (Fig. S34) (36). We observed that a parent ion for DM-IPA shows an increase of 12 mass units, indicating a formation of [¹³C₁₁, ¹⁵N]IPA in *Arabidopsis* (Fig. S3B). From the analysis of DM-IPA and [¹³C₁₁, ¹⁵N]DM-IPA, 95% of total IPA was efficiently labeled in this condition, in which 91% of total IAA was labeled (Fig. S3 C and D). This result indicated that IPA is mainly produced from Trp in *Arabidopsis*.

By using a synthetic [¹³C₁₁, ¹⁵N]IPA as an internal standard, we quantified IPA levels in *Arabidopsis*. The level of IPA in 3-wk-old WT seedlings was 53.8 ± 7.5 ng/gfw (Figs. 3A and 4A). IPA levels may vary depending on tissue type, growth stage, and environmental conditions (37). A recent study indicated that upper inflorescences produce relatively higher levels of IAA compared with other vegetative tissues in *Arabidopsis* (24). We found that the level of IPA increased by 6.9 times in the buds relative to that in WT seedlings (Figs. 3A–C and 4A and B). The endogenous level of IAA increased 5.1 times in the buds (53.6 ± 16 ng/gfw; *n* = 3) relative to that in WT seedlings (10.6 ± 1.6 ng/gfw; *n* = 3). We note that IPA levels may also vary depending on plant species, as the moss *Physcomitrella patens* gametophytes accumulate 25.0 ± 2.1 ng/gfw (*n* = 4) and maize leaves involve 39.4 ± 7.2 ng/gfw (*n* = 5) of endogenous IPA, respectively.

To investigate whether TAA1 produces IPA in vivo, we analyzed IPA levels in 3-wk-old seedlings of TAA-deficient *wei8-1 tar2-1* double mutants (Fig. 3D). The level of IPA was reduced by 32% in *wei8-1 tar2-1* compared with that in WT seedlings (Fig. 4A). We also analyzed IPA levels in the buds of *wei8-1 tar2-2*

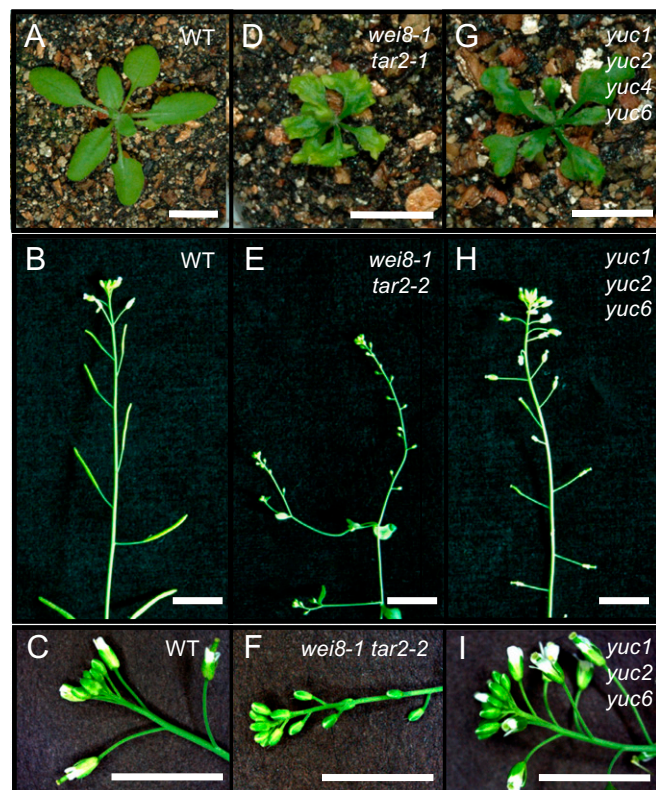


Fig. 3. Phenotypes of TAA-deficient and YUC-deficient mutants. (A) Three-week-old WT seedlings. (B) Upper region and (C) inflorescence of 7-wk-old WT plants. (D) Three-week-old seedlings of *wei8-1 tar2-1* mutants. (E) Upper region and (F) inflorescence of 7-wk-old *wei8-1 tar2-2* mutants. (G) Three-week-old seedlings of *yuc1 yuc2 yuc4 yuc6* mutants. (H) Upper region and (I) inflorescence of *yuc1 yuc2 yuc6* mutants. (Scale bars: 1 cm.)

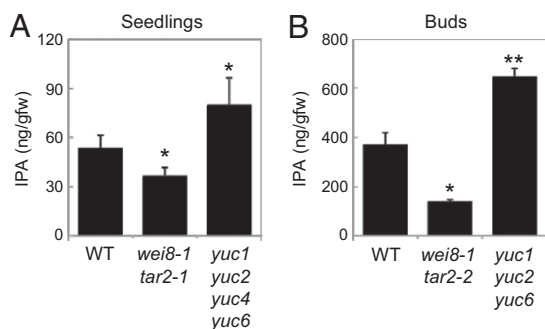


Fig. 4. The level of IPA in WT plants and TAA-deficient and YUC-deficient mutants. (A) Aerial parts of 3-wk-old seedlings grown in soil were used for IPA analysis. Values are mean $\pm$ SD ($n = 4$). (B) The buds of 7-wk-old plants before flowering were used for IPA analysis. Values are mean $\pm$ SD ($n = 3$). Differences between WT and mutants are statistically significant at $P < 0.05$ (* $P < 0.05$ and ** $P < 0.01$, t test).

double mutants, a weaker TAA-deficient mutant that is able to make flowers (Fig. 3 E and F). The level of IPA was reduced by 62% in the buds of the double mutants compared with WT plants (Fig. 4B). Moreover, we analyzed the level of IPA in *TAA1ox* (Fig. 2A). IPA levels were increased 2.9 times in *TAA1ox* relative to that in *pER8* seedlings (Table 2). These results provide in vivo evidence that TAA family plays a major role in the production of IPA in *Arabidopsis*.

YUC Catalyzes Conversion of IPA to IAA. A previous study showed that YUC1 converts TAM to HTAM in vitro (23). To investigate whether TAM metabolism is affected in YUC-deficient mutants, we analyzed TAM levels in *yuc1 yuc2 yuc4 yuc6* quadruple mutants by using $^{15}\text{N}_2$ -TAM as an internal standard (Fig. 3G). However, no significant accumulation of TAM was observed in 3-wk-old seedlings of *yuc1 yuc2 yuc4 yuc6* (209 ± 4 pg/gfw; $n = 3$) relative to that in WT seedlings (209 ± 15 pg/gfw; $n = 3$). This result suggests that YUC may not catalyze conversion of TAM to HTAM in vivo (38).

To examine whether YUC family acts in the conversion of IPA to IAA in the IPA pathway, we analyzed IPA levels in the seedlings of *yuc1 yuc2 yuc4 yuc6* quadruple mutants (Fig. 3G). We found that the level of IPA increased 1.5 times in *yuc1 yuc2 yuc4 yuc6* relative to that in WT seedlings (Fig. 4A). We further analyzed IPA levels in the buds of *yuc1 yuc2 yuc6* triple mutants, weaker alleles that form flowers (Fig. 3 H and I). Similarly, the level of IPA was increased significantly (1.8 times) in the buds of *yuc1 yuc2 yuc6* compared with that in the buds of WT (Fig. 4B). In contrast, IPA levels were 33% reduced in *YUC6ox* plants relative to that in *pER8* plants (Table 2). These results demonstrate that YUC family is most likely implicated in the conversion of IPA to IAA in *Arabidopsis* (Fig. 1B).

Table 2. IAA and IAA precursor and IAA metabolite levels in *TAA1*- and *YUC6*-overexpressing plants

Plants	IAA, IAA precursors, and IAA metabolites (ng/gfw)				
	IPA	IAAld	IAA	IAA-Asp	IAA-Glu
<i>pER8</i>	56.0 $\pm$ 8.0	11.3 $\pm$ 2.2	16.1 $\pm$ 1.0	ND	0.9 $\pm$ 0.2
<i>TAA1ox</i>	165 $\pm$ 12*	9.2 $\pm$ 0.3	19.5 $\pm$ 1.4	ND	0.5 $\pm$ 0.1*
<i>YUC6ox</i>	37.5 $\pm$ 4.0*	9.9 $\pm$ 1.9	23.7 $\pm$ 5.1	28.0 $\pm$ 5.8	10.7 $\pm$ 2.5*

Eleven-day-old seedlings were transferred to Murashige-Skoog agar media containing Est (10 μM) and grown vertically for 3 d. ND, not detected. Values are mean $\pm$ SD, $n = 3$ except for IPA ($n = 4$).

*Significantly different from *pER8* plants ($P < 0.05$, t test).

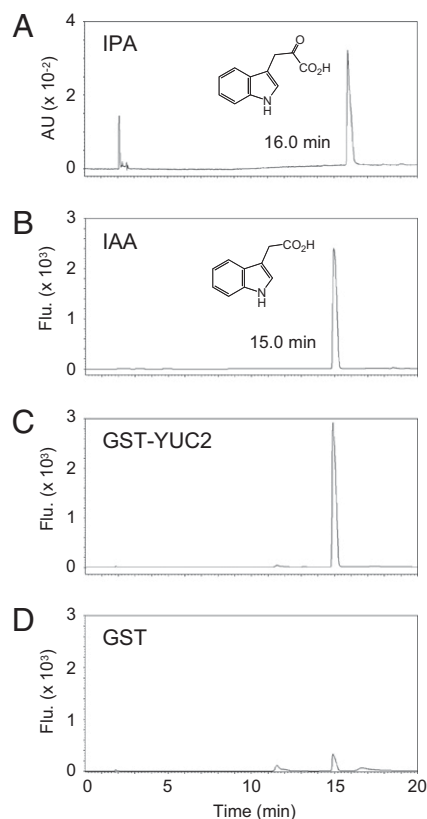


Fig. 5. Conversion of IPA to IAA by YUC2. (A) The HPLC profile for authentic IPA with UV detection (328 nm). (B) The HPLC profile for authentic IAA, (C) GST-YUC2 reaction mixture, and (D) GST reaction mixture with fluorescence detection (280 nm excitation and 355 nm emission).

To provide direct evidence that YUC catalyzes the conversion of IPA to IAA, we performed an enzyme assay by using GST-fused YUC2 (GST-YUC2) heterologously expressed in *Escherichia coli*. Purified GST-YUC2 actively converted IPA to IAA in an NADPH-dependent manner (Fig. 5 A–C and Fig. S4A). Only small amounts of IAA were produced nonenzymatically from IPA in a control reaction containing GST (Fig. 5D). The production of IAA was confirmed by LC-ESI-MS/MS (Fig. S4B). No conversion of IPA to IAAld by GST-YUC2 was observed. TAM was not a substrate of GST-YUC2 in our assay condition (Fig. S4A).

IAAld Is Probably Not Involved in IPA Pathway. Direct conversion of IPA to IAA by YUC2 protein indicates that IAAld is probably not involved in the IPA pathway. To complement our in vitro evidence, we investigated the biosynthesis pathway for IAAld in *Arabidopsis*. IAAld was previously identified in *Arabidopsis* using GC-MS (39), yet a reliable and definitive IAAld analysis method has not been established. We converted IAAld to its stable hydrazone derivative (DNPH-IAAld) in the crude extracts (Fig. S5A), and analyzed by LC-ESI-MS/MS in the negative ion mode (Fig. S5 B–I).

We tested whether IAAld is mainly produced from Trp in *Arabidopsis* by feeding a [$^{13}\text{C}_{11}$, $^{15}\text{N}_2$]Trp to *trp1-1* (Fig. S6A). We detected a parent ion for DNPH-IAAld with increase of 11 mass units, suggesting a formation of [$^{13}\text{C}_{10}$, ^{15}N]IAAld in *Arabidopsis* (Fig. S6B). Analysis of ^{13}C and ^{15}N -incorporation rate indicates that 99% of total IAAld was labeled under this condition, in which 91% of total IAA was labeled (Fig. S6C). This result indicated that IAAld is mainly produced from Trp in *Arabidopsis*.

By using a synthetic [$^{13}\text{C}_{10}$, ^{15}N]IAAld as an internal standard, the level of IAAld was quantified as 15.1 ± 5.3 ng/gfw ($n = 3$) in 2-wk-old WT seedlings of *Arabidopsis*. Although the IPA levels were increased drastically in *TAA1ox* plants, IAAld levels did not show a significant change relative to that in *pER8* (Table 2). We further observed that IAAld levels were not reduced, but rather increased, in the buds of *wei8-1 tar2-2* mutants (33.9 ± 3.9 ng/gfw; $n = 2$) compared with WT (23.8 ± 1.7 ng/gfw; $n = 2$), in which IPA levels were reduced (Fig. 4B). Moreover, IAAld levels were not affected in *YUC6ox*, in which IAA-amino acid conjugate levels were significantly increased (Table 2). These observations indicate that IAAld is most likely not implicated in the IPA pathway, but in another Trp-dependent pathway.

We examined whether the AO family is involved in IAA biosynthesis by analyzing IAAld levels in *aba3* mutants, in which all AO members are inactivated. IAAld levels would be increased if the AO family were implicated in the oxidation of IAAld in plants. However, no increase of IAAld levels was observed in *aba3* mutants (15.0 ± 2.5 ng/gfw; $n = 3$) compared with that in WT plants (15.1 ± 5.3 ng/gfw; $n = 3$), in which IAA and IAA-Glu levels were also not significantly changed (Fig. S7). This result indicates that the AO gene family probably does not play a role in IAA biosynthesis.

Discussion

We provide multiple lines of evidence that the TAA family produces IPA and the YUC family catalyzes the conversion of IPA to IAA in *Arabidopsis*. TAA and/or YUC families play critical roles in embryogenesis, flower development, seedling growth, vascular patterning, lateral root formation, tropism, shade avoidance, and temperature-dependent hypocotyl elongation (19, 20, 25–27). Thus, we conclude that the IPA pathway is the major IAA biosynthesis pathway in *Arabidopsis*. The YUC family mediates a rate-limiting step in the IPA pathway. *TAA1* and *YUC* can act synergistically to enhance IAA biosynthesis in *Arabidopsis* (Table 1). The expression patterns of *TAA* and *YUC* families are spatiotemporally regulated in plant development (19, 20, 25–27). These results indicate that *TAA* and *YUC* families may coordinately regulate IAA production. Further analysis of the expression patterns of *TAA* and *YUC* families would be a key to understanding the sites and regulation of IPA-dependent IAA biosynthesis in plants.

YUC2 protein catalyzes the direct conversion of IPA to IAA. YUC proteins may function similarly to lactate monooxygenases that convert lactate to acetic acid and CO_2 via pyruvate (40). Further kinetic and structural analyses of YUC proteins would clarify the molecular mechanism of IAA formation. IAAld has been proposed as an intermediate of the IPA pathway, but may be in another pathway in *Arabidopsis*. A recent study suggests

that IAAld is an IAA precursor produced from TAM in the pea (Fig. 1B) (14). Quittenden et al. demonstrated that D₅-TAM was incorporated to IAAld in pea roots by using GC-MS. TAM and IAAld have been detected in *Arabidopsis* and pea (9, 14), but genetic evidence has not been provided for the occurrence of the TAM pathway in plants. Trp DECARBOXYLASE (TDC) that catalyzes the conversion of Trp to TAM has been cloned and characterized in some plant species (41, 42). However, TDC genes have not been identified in *Arabidopsis*. The AO family members have been demonstrated to oxidize IAAld to IAA in vitro, but our results show that AO is probably not involved in IAA biosynthesis in *Arabidopsis*. Thus, the TAM pathway may operate in the pea, but it is not clear whether this pathway also exists in other plants.

Materials and Methods

Plant Materials and Growth Conditions. *Arabidopsis thaliana* ecotype Columbia-0 was used as the WT control. Transgenic plants used in this study are described in *SI Materials and Methods*. *yuc1 yuc2 yuc6* and *yuc1 yuc2 yuc4 yuc6* were generated from *yuc1*–*yuc2*+/+ *yuc4*–*yuc6*– plants, *wei8-1 tar2-1* from *wei8-1*–*tar2-1*+/+, and *wei8-1 tar2-2* from *wei8-1*–*tar2-2*+/+ (19, 26). The *trp1-1* and *aba3-1* mutants were obtained from the *Arabidopsis* Biological Resource Center (ABRC). After imbibitions at 4 °C for 2 d, surface-sterilized seeds were germinated on Murashige–Skoog agar media (pH 5.7) supplemented with thiamin hydrochloride (3 µg/mL), nicotinic acid (5 µg/mL), pyridoxine hydrochloride (0.5 µg/mL), myoinositol (100 µg/mL), 1% (wt/vol) sucrose, and 0.8% agar. Plants were grown at 21 °C under continuous white light (30–50 µmol·m^{−2}·s^{−1}). When grown on soil, 2-wk-old seedlings were transferred to soil and cultivated in a temperature-controlled chamber.

Chemical Synthesis, LC-ESI-MS/MS, Labeling Experiments, and Enzyme Assay. [$^{13}\text{C}_{11}$, ^{15}N]IPA, [$^{13}\text{C}_{10}$, ^{15}N]IAAld, [$^{13}\text{C}_4$, ^{15}N]IAA-Asp, and [$^{13}\text{C}_5$, ^{15}N]IAA-Glu were synthesized as described in *SI Materials and Methods*. LC-ESI-MS/MS analysis of IAA and IAA precursors, in vivo labeling experiments, and YUC enzyme assay were performed as described in *SI Materials and Methods* and Table S1.

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